Levi Neuwirth

RESEARCH ENGINEER — TECHNICAL AI ASSURANCE

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Research engineer working on verifiable inference and frontier-model evaluations. Combines cryptographic verifier hardening and GPU workload profiling with evaluator calibration, agent failure analysis and reproducible experiments.

RESEARCH & ENGINEERING EXPERIENCE

Cambridge AI Safety Hub — *MARS V Fellow*

July 2026 – Present · Oxford, United Kingdom

VerInf: zero-knowledge LLM inference research with James Petrie (Future of Life Institute). Cited as prior work by Anthropic (November 2026).

- Built and upstreamed a Python workload profiler, versioned manifests, execution DAGs and cost models; validated workload accounting against an archived Llama-4 Maverick run and calibrated on NVIDIA B200 hardware.
- Implemented verifier-transparent weight splitting: partitioned proofs matched the ordinary path byte for byte and passed the Rust verifier on one GPU. Multi-device speedup has not been measured.
- Integrated a collaborator's expert-weight bridge; fixed enrollment-identity binding, Merkle-index checks and sumcheck-round enforcement in Python and Rust verification paths.
- Implemented targeted causal softmax, SiLU and surprisal repairs from protocol-review findings, with a 10-case GPU regression suite. Earlier full-scale proofs do not verify under the repaired constraints.
- Identified unbound parameters in Maverick's inference driver; implemented enrollment repairs awaiting GPU validation. Hardened the test harness to distinguish explicit Rust verifier verdicts from crashes or missing output.

VerInf · Case study

Independent Research Contractor — *Frontier AI labs*

2025 – Present

Lead frontier-model evaluation-environment design and evaluator calibration.

- Build tasks and scoring rubrics in **Harbor**; execute models and debug agent scaffolds and trajectories.
- Identify rubrics that measure something other than the intended task; refine evaluation design before scaling runs. Distinguish model failures from defects in task design, scaffolding and scoring.
- Analyze reasoning, capability, safety and red-team failures. Client identities and specific results are confidential.

Proof Broker — Independent Research *2026 – Present*. Built an OCaml broker connecting Lean 4 and Rocq to untrusted solvers and model-generated search through versioned evidence schemas, independent certificate checks and kernel-checked proof terms.

- Built reproducible Python evaluation workflows with pinned tasks, deterministic baselines, negative controls and replayable records; evaluated live-model certificate proposals on 15 fixed obligations.
- Audited context dependencies in kernel-accepted proofs; repaired integer-goal reconstruction, the final normalizer and audit parameter binding to test whether certificates alone justified the result.
- Offline replay produced kernel-checked proofs from 36/36 retained certificates at four obligations with no added axioms.

Shu Laboratory, Brown University Department of Neurology — *Undergraduate Researcher and Technical Lead*

October 2023 – May 2026 · Providence, RI

- Co-led Deep Sets outcome models developed on an 80M+ discharge cohort; readmission AUROC was 0.750 versus 0.655 for the Charlson index on 3.23M temporally held-out discharges. Deployed the public calculator with a Python/FastAPI backend; manuscript under review at JAMIA.
- Technical lead for NeuroPose across four externally funded projects: Python/TensorFlow inference, validated outputs, persistent job state across crashes, Docker deployment and Rust-backed analysis.

TECHNICAL SKILLS

Languages: Python, Rust, C, C++, Go.
Evaluation: Harbor, Evaluation-environment design, Evaluator calibration, Agent scaffolds, Red-teaming.
ML and data: PyTorch, TensorFlow, NumPy, Pandas.
Verification: Lean 4, Rocq, OCaml, SMT (cvc4, cvc5, Z3), Zero-knowledge proofs.
Platforms (Linux): Git, Docker, SLURM.

SELECTED EVALUATION & SYSTEMS PROJECTS

The Specification Dilemma *2026*. Designed and ran a preregistered experiment on prompt specificity and LLM output diversity using 30 matched prompt pairs. Built generation and embedding-analysis pipelines; diagnosed ignored seeds, reasoning-budget effects, and empty completions. Published code, outputs, and limitations; a single-model probe using authored user scenarios.

LeVCS *April 2026 – Present*. Built a federated version control system in Rust with signed authority chains and three storage modes. Implemented journaled transactions, group commit, committed snapshots, idempotent retries and crash recovery.

Pmacs *2025 – Present*. Built a Rust-cored, Lua-scripted editor with a long-lived runtime and thin terminal/GPU frontends over a typed protocol. Implemented process supervision, reconnectable sessions and CRDT-backed shared buffers.

Weenix *January 2025 – August 2025*. Built and extended a 10,000-line Unix-like kernel in C: virtual memory, VFS and file systems, system calls, threads, devices and interrupts, userspace x86-64 ELF support, pipes, and userspace preemption.

Post-Quantum Cryptography / SIMD *March 2025 – Present*. Benchmarked four ML-KEM / Kyber build variants on x86-64 AVX2, separating compiler optimization, auto-vectorization and hand-written SIMD. Measured 35–56× core NTT and 5.4–7.1× end-to-end speedups over optimized portable C, with statistical analysis and a reproducible artifact.

EDUCATION

Brown University — *Sc.B., Mathematics and Computer Science*
August 2022 – May 2026 · Providence, RI

GPA 3.8/4.0. Individual research grant: Brown Undergraduate Teaching and Research Award (UTRA), Summer 2024.

SELECTED RESEARCH & PRESENTATIONS

Shu L, Neuwirth L†, Wang X†, Zheng H†. *Beyond Comorbidity Indices: An Order-Invariant ICD-10-CM Embedding for Readmission and Mortality Prediction.* Under review at *Journal of the American Medical Informatics Association* (JAMIA), 2026. † Equal contribution among undergraduate authors.

Neuwirth L, Dasher AS. *Proving What a Datacenter Did: Verified Inference Between Adversaries.* MARS V, Cambridge AI Safety Hub, July 2026. Presented.

Neuwirth L. *Timed Guarding in Degree-Reduction Clouds.* Preprint, 2026, September.

Neuwirth L. *Branch-Tube Persistence and Static Coverage in Tree-Ball Geometry.* Preprint, 2026, July.

Neuwirth L. *The Annealed Critical Window for Growing-Radius Domination in Random Regular Graphs.* Preprint, 2026, July.